

Pinning group consensus of multi-agent systems under DoS attacks

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Abstract

In this paper, group consensus is investigated for a class of nonlinear multi-agent systems suffered from the DoS attacks. Firstly, a first-order nonlinear multi-agent system is constructed, which is divided into M subsystems and each subsystem has an unique leader. Then a protocol is proposed and a Lyapunov function candidate is chosen. By means of the stability theory, a sufficient criterion, which involves the duration of DoS attacks, coupling strength and control gain, is obtained for achieving group consensus in first-order system. That is, the nodes in each subsystem can track the leader of that group. Furthermore, the result is extended to nonlinear second-order multi-agent systems and the controller is also improved to obtain sufficient conditions for group consensus. Additionally, the lower bounds of the coupling strength and average interval of DoS attacks can be determined from the obtained sufficient conditions. Finally, several numerical simulations are presented to explain the effectiveness of the proposed controllers and the derived theoretical results.

Keywords: multi-agent system, group consensus, DoS attack, Lyapunov function.

1 Introduction

With the rapid development of information technology and network science, multi-agent systems (MASs for short) are gradually applied in numerous aspects, such as autonomous unmanned vehicles[1], unmanned aerial vehicle formation[2], factory automation management[3], etc. Due to the large scale of the MASs, however, it is more difficult for such systems to return to normal state timely when they suffer from network attacks.

Common network attacks include denial of service (DoS for short) attacks[4], distributed denial of service (DDoS for short) attacks[5], false data injection (FDI for short) attacks[6], etc. Among them, DoS attacks are relatively destructive. The working principle of DoS attacks is to compel the networked system too busy to process necessary instructions and occupy the key communication resources. DoS attacks will use packets to overwhelm the local system to disturb or intensely stop the corresponding local services answering the reasonable external demand, and sometimes will crumble the local system so that it doesn't work properly.

To deal with such attacks, the system needs to have the ability to recover within a limited time after being attacked to ensure that the system achieves the required collective dynamics. Consensus is a typical collective dynamics and a basic research topic in the distributed cooperative coordination of MASs. It refers to the fact that all agents are coupled into a network only based on local information interaction, and a suitable distributed cooperative controller needs to be designed so that each agent eventually reach an identical state [7]. Various studies are conducted on coordination of MASs without or with DoS attacks. Specifically, in [8], a nonlinear MAS is investigated under switching networked topology, and a consensus protocol with the relative output feedback is designed. In [9], based on stochastic and nonlinear stability theory, a finite mean square consensus criterion is established for a second-order nonlinear MAS. In [10], consensus of MASs is realized via the proposed event-triggered control strategy without collecting global information. In [11], a second-order MAS in directed networks is studied, and practical consensus is reached in a fixed time. In [12], the security control problem of MASs under DoS attacks is studied, impulsive protocol is proposed to achieve consensus. In [13], multiple channels are affected by DoS attacks during communication, event-triggered-based distributed controllers are proposed to reach bipartite consensus. In [14], based on the distributed control, group consensus is achieved for the first-order MASs under DoS attacks and switching topologies. Therefore, it is significant to further explore the impact of DoS attacks on achieving consensus. In particular, the impact of DoS attacks on other collective dynamics needs to be further discussed.

When a MAS reaches consensus, all nodes of the system tend to a common state. Therefore, a common target or a leader is beneficial to analysis consensus. In [15], the necessary and sufficient criteria are derived to guarantee bipartite consensus in the sense of mean square between each agent and objective. In [16], event-based and self-triggered strategies are designed to handle with the impact of DoS attacks in MASs, and leader-following consensus is further realized. One can see that the design of appropriate controls plays a very significant role for achieving consensus in the system, especially for the system suffered from DoS attacks. In [17], event-trigger-based leader-following bipartite consensus is studied in the nonlinear MASs via the event-triggered control, in addition, the leader with non-zero input is further introduced. In [18], a nonlinear signed network is studied, bipartite leader-follower consensus is reached via the impulsive control. All in all, it can be seen that the leader plays an important reference role in achieving consensus.

Different from the MAS with one leader, when the MAS has multiple isolated leaders, all nodes of the MAS tend to reach group consensus. Therefore, group consensus is also studied extensively. In [19], by utilizing the discontinuous stability theory, group consensus is studied over both fixed topology and Markov switching topology. In [20], by applying frequency domain analysis method and matrix analysis theory, the sufficient criteria for the MAS to achieve group consensus are obtained. In [21], group consensus of a heterogeneous MAS is investigated, and two control schemes are presented to reach group consensus of the system with uncertain parameters. Up to now, studies on group consensus of MASs suffered from DoS attacks are relatively rare. Especially under DoS attacks, the problem of group consensus of the MASs with multiple isolated leaders deserves further study.

Enlightened by the above discussions, this article mainly focuses on group consensus of the MAS with multiple isolated leaders. Main contribution of this article are highlighted below.

(1) For a class of nonlinear MASs with multiple isolated leaders, group consensus of the nonlinear MASs is analyzed, and sufficient criteria for ensuring group consensus are derived.

(2) DoS attacks, which can be conducted aperiodically or randomly, are considered in information interaction among agents. The effect of attack durations on consensus is clearly given.

(3) The group consensus problem for a first-order nonlinear MAS is first studied under DoS attacks, and then the theoretical results are further generalized to the second-order nonlinear MAS.

2 Preliminaries

We suppose that the MASs have M leader agents and N follower agents. These followers can be divided into M groups, and each group has an unique leader, $N = \sum_{j=1}^M N_j$, where N_j denotes the total number of follower nodes in group j , $j = 1, 2, \dots, M$.

Define the set $I_0 = \{N+1, \dots, N+M\}$, $I_1 = \{1, 2, \dots, N_1\}$, $I_k = \{\sum_{j=1}^{k-1} N_j + 1, \sum_{j=1}^{k-1} N_j + 2, \dots, \sum_{j=1}^{k-1} N_j + N_1\}$, $k = 2, 3, \dots, M$, and $I = I_1 \cup I_2 \cup \dots \cup I_M$. The node set is $V_j = \{n_i \mid i \in I_j\}$, $j = 0, 1, 2, \dots, M$, where V_0 is the node set of the leaders, and the rest sets are the follower nodes.

The triple $G = (V, E, A)$ represents a weighted directed graph and also indicates the relationship among those followers. The set of followers is V , which has a partition $V = V_1 \cup V_2 \cup \dots \cup V_M$. The edge set is $E \subseteq V \times V$. In addition, the adjacency matrix is $A = [a_{ij}]_{N \times N}$, where $a_{ij} \neq 0$ if there is a direct edge from node j to node i , otherwise $a_{ij} = 0$.

The set of neighbours for node n_i is defined as:

$$N_{k,i} = \{n_j \in V_k \mid (n_i, n_j) \in E\}, k = 1, 2, \dots, M.$$

Observing the information interaction between two agents in the same subsystem and also between two agents in the different subsystems, this paper divides the agents as follows [22]: $V_k^{in} = \{n_i \in V_k \mid \bigcup_{l \neq k} N_{l,i} = \emptyset \wedge n_i \notin \bigcup_{m \in I \setminus I_k} N_{k,m}\}$ is the k -th node

set that does not exchange information with the nodes in other subsystems, and $V_k^{out} = \{n_i \in V_k \mid \bigcup_{l \neq k} N_{l,i} \neq \emptyset \vee n_i \in \bigcup_{m \in I \setminus I_k} N_{k,m}\}$ is the k -th node set that exchanges information with the nodes in the k -th subsystem and other subsystems. According to the above definition, we can obtain $V_k = V_k^{in} \cup V_k^{out}$ and $V_k^{in} \cap V_k^{out} = \emptyset$. Define $V^{in} = \bigcup_{k=1}^M V_k^{in}$ and $V^{out} = \bigcup_{k=1}^M V_k^{out}$.

Furthermore, let $L = [l_{ij}]_{N \times N}$ be the Laplacian matrix of G with $l_{ii} = \sum_j^N a_{ij}$ and $l_{ij} = -a_{ij}$ for $i \neq j$.

Assumption 1. *The subgraph of each group exists a spanning tree with one leader agent as its root.*

DoS attack is one of the most common way of the network attack, which compels the MASs too busy to process necessary instructions and occupies the key communication resources. Therefore, when the MASs are subjected to the DoS attack, the MASs cannot execute the designed control instructions, in other words, the control fails during the attack period. In this paper, an aperiodic DoS attack model is established. For the sake of storing energy for the next attack, the DoS attack needs to retain in a dormant state for a while after each attack. The specific description of the DoS attack of this paper is introduced below.

Let the DoS attack sequence be $\{t_\theta\}, \theta \in N^*, N^* = \{0, 1, 2, \dots\}$. Then the DoS attack interval can be expressed as $S_\theta = [t_\theta, t_\theta + \tau_\theta)$, where τ_θ is the duration of the DoS attack. The intersection of all intervals where DoS attacks occur is an empty set, that is, there is no overlap, which can be expressed as $t_\theta + \tau_\theta < t_{\theta+1}, \theta \in N^*$. For all given bounded time interval $[t_0, t], t_0 < t$, it is impossible to transmit information during the DoS attack time. These time interval that cannot transmit information can be expressed as $T_D = \bigcup S_\theta \cap [t_0, t]$. In that case, the corresponding time interval for information transmission can be expressed as $T_s = [t_0, t] \setminus \bigcup S_\theta$.

Definition 1. [23] *The DoS attack frequency is defined as $F_s(t_0, t) = \frac{N_s(t_0, t)}{t - t_0}$, where $N_s(t_0, t)$ is the number of attacks during the interval $[t_0, t]$.*

Assumption 2. *With regard to the duration of DoS attack, there exist $T > 1$ and $\varphi > 0$ such that $|T_D| < \varphi + \frac{t - t_0}{T}$, where T and φ are both constants, and $|T_D|$ represents the time length of T_D .*

Remark 1. *In the DoS attack interval, the communications between two adjacent agents will fail properly. This paper aims to study group consensus of the MASs suffered from DoS attacks and design a suitable controller so that followers in each subgroup can still approach the state information of the corresponding leader in that subgroup.*

3 Main results on group consensus

In this section, group consensus of first-order MASs is first analyzed, and then the result is further extended to the second-order MASs. Several sufficient conditions are obtained.

3.1 Group consensus of first-order MASs under DoS attacks

The leaders are modeled as

$$\dot{x}_k^*(t) = f(x_k^*(t), t), \quad k \in I_0, \quad (1)$$

and the follower agents are represented as

$$\dot{x}_i(t) = f(x_i(t), t) + \mu_i(t), \quad i \in I, \quad (2)$$

where $x_k^*(t), x_i(t), f(x_k^*(t), t), f(x_i(t), t) \in R^n$, and $\mu_i(t) \in R^n$ is external controller to be designed.

Definition 2. *If first-order nonlinear MASs (1)-(2) are in accord with $\lim_{t \rightarrow \infty} \|x_i(t) - x_k^*(t)\| = 0$, $i \in I_k$ under any initial conditions, then the group consensus of MASs (1)-(2) under DoS attacks is said to be realized.*

Assumption 3. *For arbitrary variable $w_1, w_2 \in R^n$, there is a non-negative real number ρ such that the nonlinear function satisfies*

$$\|f(w_1, t) - f(w_2, t)\| \leq \rho \|w_1 - w_2\|.$$

The following rules are made for information interaction.

(i) When $n_i \in V^{in}(i \in I)$ are the nodes whose leader's state of the subsystem is known, and the information interaction in this subsystem is transmitted by the position state $x_i(t)$, and the error state is expressed as $e_i = x_i - x_i^*$, where $x_i^* = x_k^*$ for $i \in I_k, k = 1, 2, \dots, M$.

(ii) When $n_i \in V^{out}(i \in I)$ are the nodes whose leader's state of the subsystem is known, the state of the out-group neighbors are known and the leader's state of those out-group neighbors are unknown, the information interaction in this subsystem is transmitted by the position state $x_i(t)$ and the out-group information exchange is transmitted by the error of the position state $x_i - x_i^*$, where $x_i^* = x_k^*$ for $i \in I_k, k = 1, 2, \dots, M$.

Next, we design the external controller for first-order MASs as

$$\begin{aligned} \mu_i(t) = \alpha c \left[\sum_{j \in V_k} a_{ij} (x_j(t) - x_i(t)) + b_i (x_k^*(t) - x_i(t)) \right. \\ \left. + \sum_{j \in (V \setminus V_k)} a_{ij} (e_j(t) - e_i(t)) \right], \quad i \in V_k, \end{aligned} \quad (3)$$

where c represents the controller gain and α indicates the coupling strength. When the MASs suffer from DoS attack, $c = 0$. Accordingly, when there are no attacks, $c > 0$. a_{ij} refers to the weight for the edge between the i th node and the j th node. Additionally, b_i refers to pinning coupling strength of the i th node and its leader. We define matrix $B = \text{diag}\{b_1, b_2, \dots, b_N\}$, $H = [h_{ij}]_{N \times N}$, $H = L + B$, $H' = \frac{H + H^T}{2}$.

Based on the properties of the Laplacian matrix, the external controller (3) can also be transformed to

$$\begin{aligned}
\mu_i(t) &= \alpha c \left[\sum_{j \in V_k} a_{ij} (x_j(t) - x_k^*(t)) - \left(\sum_{j \in V_k} a_{ij} + b_i \right) (x_i(t) - x_k^*(t)) \right. \\
&\quad \left. + \sum_{j \in V \setminus V_k} a_{ij} e_j(t) - \sum_{j \in V \setminus V_k} a_{ij} e_i(t) \right] \\
&= \alpha c \left[\sum_{j \in V_k} a_{ij} e_j(t) - \left(\sum_{j \in V_k} a_{ij} + \sum_{j \in V \setminus V_k} a_{ij} + b_i \right) e_i(t) + \sum_{j \in V \setminus V_k} a_{ij} e_j(t) \right] \\
&= \alpha c \left[\sum_{j \in V_k, i \neq j} -l_{ij} e_j(t) - (l_{ii} + b_i) e_i(t) - \sum_{j \in V \setminus V_k} l_{ij} e_j(t) \right] \\
&= -\alpha c \left[\sum_{j \in V} l_{ij} e_j(t) + b_i e_i(t) \right] \\
&= -\alpha c \sum_{j \in V} h_{ij} e_j(t),
\end{aligned}$$

therefore, for $t \in [t_\theta + \tau_\theta, t_{\theta+1})$, $\theta \in N^*$, DoS attack is dormant and the error system of the MASs can be expressed as

$$\begin{aligned}
\dot{e}_i(t) &= \dot{x}_i(t) - \dot{x}_k^*(t) \\
&= -\alpha c \sum_{j \in V} h_{ij} e_j(t) + [f(x_i(t), t) - f(x_k^*(t), t)].
\end{aligned} \tag{4}$$

Reformulate the error systems (4) into a compact form

$$\dot{e}(t) = -\alpha c (L + B) \otimes I_n e(t) + F(x, t) = -\alpha c H \otimes I_n e(t) + F(t), \tag{5}$$

where $e(t) = [e_1^T(t), e_2^T(t), \dots, e_N^T(t)]^T$ and $F(t) = [F_1^T(t), F_2^T(t), \dots, F_N^T(t)]^T$ with $F_i(t) = f(x_i(t), t) - f(x_k^*(t), t)$, $i = 1, 2, \dots, N$.

When $t \in [t_\theta, t_\theta + \tau_\theta)$, $\theta \in N^*$, DoS attack is active and the external control is failing, that is $c = 0$. Then, the error system of the MASs can be expressed as

$$\dot{e}(t) = F(t). \tag{6}$$

In fact, system (6) can be written uniformly to system (5), which becomes system (6) if $c = 0$. Next, we present the theoretical result on guaranteeing group consensus of the first-order MASs (1) and (2) suffered from the DoS attacks via the distributed cooperative controller (3).

Theorem 1. *When the assumptions 1 and 3 are true, the first-order MASs (1) and (2) under the DoS attacks can achieve group consensus via the distributed cooperative controller (3) if the following conditions hold:*

$$\begin{aligned}
(a_1) \quad & c > \frac{\rho^2 + 1}{2\alpha\lambda_{\min}(H')}; \\
(b_1) \quad & T > \frac{2\alpha c\lambda_{\min}(H')}{2\alpha c\lambda_{\min}(H') - \rho^2 - 1}.
\end{aligned}$$

Proof. Lyapunov function is designed as

$$V(t) = \frac{1}{2} e^T(t) e(t).$$

Then, computing the derivative of $V(t)$ derives

$$\begin{aligned} \dot{V}(t) &= e^T(t) \dot{e}(t) \\ &= e^T(t) (-\alpha c H \otimes I_n e(t)) + e^T(t) F(t) \\ &\leq e^T(t) (-\alpha c H' \otimes I_n e(t)) + \frac{1}{2} e^T(t) e(t) + \frac{1}{2} F^T(t) F(t) \\ &\leq e^T(t) (-\alpha c \lambda_{\min}(H') e(t)) + \frac{1}{2} e^T(t) e(t) + \frac{1}{2} \rho^2 e^T(t) e(t) \\ &= e^T(t) \left(-\alpha c \lambda_{\min}(H') + \frac{1}{2} + \frac{1}{2} \rho^2 \right) e(t) \\ &= 2 \left(-\alpha c \lambda_{\min}(H') + \frac{1}{2} + \frac{1}{2} \rho^2 \right) V(t) \\ &= (-2\alpha c \lambda_{\min}(H') + 1 + \rho^2) V(t). \end{aligned}$$

When $t \in [t_\theta, t_\theta + \tau_\theta]$, DoS attacks occur, that is, $c = 0$, therefore,

$$\dot{V}(t) \leq (1 + \rho^2) V(t), t \in [t_\theta, t_\theta + \tau_\theta].$$

When $t \in [t_\theta + \tau_\theta, t_{\theta+1}]$, $c > 0$, hence,

$$\dot{V}(t) \leq (-2\alpha c \lambda_{\min}(H') + 1 + \rho^2) V(t), t \in [t_\theta + \tau_\theta, t_{\theta+1}].$$

Therefore,

$$\begin{aligned} \dot{V}(t) &\leq V(t_0) e^{(-2\alpha c \lambda_{\min}(H') + 1 + \rho^2)(t - t_0) + (2\alpha c \lambda_{\min}(H'))(\varphi + \frac{t - t_0}{T})} \\ &= e^{2\alpha c \lambda_{\min}(H') \varphi} e^{(-2\alpha c \lambda_{\min}(H') + 1 + \rho^2 + \frac{2\alpha c \lambda_{\min}(H')}{T})(t - t_0)}. \end{aligned}$$

According to conditions (a_1) and (b_1) ,

$$-2\alpha c \lambda_{\min}(H') + 1 + \rho^2 + \frac{2\alpha c \lambda_{\min}(H')}{T} < 0,$$

consequently, it follows from Lyapunov stability theory that $\lim_{t \rightarrow \infty} V(t) = 0$, that is $\lim_{t \rightarrow 0} \|e(t)\| = 0$.

In a word, group consensus of the MASs (1) and (2) is reached, and the theorem is proved. $\square$

Remark 2. According to Theorem 1, one knows that group consensus has a requirement on the duration of the DoS attacks. In addition, when the controller gain and the attack duration satisfy the conditions (a_1) and (b_1) respectively, group consensus of the MASs (1) and (2) can still be reached based on distributed protocol (3).

3.2 Grouping consensus of second-order MASs under DoS attacks

The leaders dynamics are given as

$$\begin{cases} \dot{x}_k^*(t) = v_k^*(t), \\ \dot{v}_k^*(t) = f(x_k^*, v_k^*, t), \end{cases} \quad k \in I_0, \quad (7)$$

and the followers dynamics are modeled as

$$\begin{cases} \dot{x}_i(t) = v_i(t), \\ \dot{v}_i(t) = f(x_i(t), v_i(t), t) + \mu_i(t), \end{cases} \quad i \in I, \quad (8)$$

where $x_k^*(t)$, $x_i(t)$, $v_k^*(t)$, $v_i(t)$, $f(x_k^*(t), v_k^*(t), t)$, $f(x_i(t), v_i(t), t) \in R^n$, and $\mu_i(t) \in R^n$ is external controller to be designed.

Definition 3. *If second-order nonlinear MASs (7) and (8) are in accord with the formulation $\lim_{t \rightarrow \infty} \|x_i(t) - x_k^*(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|v_i(t) - v_k^*(t)\| = 0$ under any initial conditions for any $i \in I_k$, then group consensus of MASs (7) and (8) under DoS attacks is said to be realized.*

Assumption 4. *For any variable $z_1, z_2, w_1, w_2 \in R^n$, there are real numbers $\rho_1 \geq 0$ and $\rho_2 \geq 0$ such that*

$$\|f(z_1, z_2, t) - f(w_1, w_2, t)\| \leq \rho_1 \|z_1 - w_1\| + \rho_2 \|z_2 - w_2\|.$$

The following rules are made for information interaction.

(i) When $n_i \in V^{in}(i \in I)$ are the nodes whose leader's state of the subsystem is known, and the information interaction in this subsystem is transmitted by the position state $x_i(t)$ and the velocity state $v_i(t)$. Additionally, the error states are expressed as $e_i^{(1)} = x_i - x_i^*$ and $e_i^{(2)} = v_i - v_i^*$, where $x_i^* = x_k^*$ and $v_i^* = v_k^*$, $i \in I_k$, $k = 1, 2, \dots, M$.

(ii) When $n_i \in V^{out}(i \in I)$ are the nodes whose leader's state of the subsystem is known, the state of the out-group neighbors are known and the leader's state of those out-group neighbors are unknown, the information interaction in this subsystem is transmitted by the position state $x_i(t)$ and the velocity state $v_i(t)$ and the out-group information exchange is transmitted by the errors of the position state $e_i^{(1)} = x_i - x_i^*$ and the velocity state $e_i^{(2)} = v_i - v_i^*$, where $x_i^* = x_k^*$ and $v_i^* = v_k^*$, $i \in I_k$, $k = 1, 2, \dots, M$.

The second-order cooperative controller is proposed as

$$\begin{aligned} \mu_i(t) = & \alpha c \left[\sum_{j \in V_k} a_{ij} (x_j(t) - x_i(t)) + b_i (x_k^*(t) - x_i(t)) + \right. \\ & \left. \sum_{j \in (V \setminus V_k)} a_{ij} (e_j^{(1)}(t) - e_i^{(1)}(t)) \right] \\ & + \beta c \left[\sum_{j \in V_k} a_{ij} (v_j(t) - v_i(t)) + b_i (v_k^*(t) - v_i(t)) + \right. \\ & \left. \sum_{j \in (V \setminus V_k)} a_{ij} (e_j^{(2)}(t) - e_i^{(2)}(t)) \right], \end{aligned} \quad (9)$$

where c refers to the control gain, and α and β signify the coupling strengths. $c = 0$ when the MASs are under DoS attack, and $c > 0$ when no attack occurs. a_{ij} means the weight for the edge of the i th node and the j th node. Furthermore, b_i denotes the pinning coupling strength of the i th node and its leader.

Represent the cooperative controller as follows

$$\mu_i(t) = -\alpha c \sum_{j \in V} h_{ij} e_j^{(1)}(t) - \beta c \sum_{j \in V} h_{ij} e_j^{(2)}(t).$$

Let

$$\begin{aligned} e^{(1)}(t)^T &= [e_1^{(1)}(t)^T, e_2^{(1)}(t)^T, \dots, e_N^{(1)}(t)^T]^T, \\ e^{(2)}(t)^T &= [e_1^{(2)}(t)^T, e_2^{(2)}(t)^T, \dots, e_N^{(2)}(t)^T]^T, \end{aligned}$$

and

$$\bar{e} = \begin{bmatrix} e^{(1)} \\ e^{(2)} \end{bmatrix}. \quad (10)$$

Then error systems of systems (7) and (8) can further be reexpressed as the following matrix equation

$$\dot{\bar{e}}(t) = \begin{bmatrix} 0_N & I_N \\ -\alpha c H & -\beta c H \end{bmatrix} \otimes I_n \bar{e}(t) + \begin{bmatrix} 0_{nN \times 1} \\ [f(x_i(t), t) - f(x_k^*(t), t)]_{N \times 1} \end{bmatrix}. \quad (11)$$

Let $M = \begin{bmatrix} 0_N & I_N \\ -\alpha c H & -\beta c H \end{bmatrix} \otimes I_n$, and $\bar{F}(t) = \begin{bmatrix} 0_{nN \times 1} \\ F(t) \end{bmatrix}$, the system is transformed into

$$\dot{\bar{e}}(t) = M \bar{e}(t) + \bar{F}(t). \quad (12)$$

Similar to subsection 3.1, system (12) can represent both an error system that is not affected by the DoS attacks and an error system that is affected by the DoS attacks. In other words, if $c = 0$ in matrix M , then system (12) represents error system that is not affected by the DoS attacks; if $c \neq 0$ in matrix M , then system (12) represents error system that is affected by the DoS attacks.

Theorem 2. Under assumptions 2 and 4, and a DoS attack in the communication, the second-order MASs composed of (7) and (8) can reach group consensus, with the distributed cooperative controller (9) if:

$$\begin{aligned} (a_2) \quad & c > \max \left\{ \frac{\alpha}{2\beta^2 \lambda_{\min}(H')}, \frac{\alpha + 2\alpha\rho_1^2 + \beta\rho_1^2}{2\alpha^2 \lambda_{\min}(H')}, \frac{2\alpha + \beta + 2\alpha\rho_2^2 + 2\beta\rho_2^2}{2\beta^2 \lambda_{\min}(H')} \right\}; \\ (b_2) \quad & T > \frac{r_S - r_D}{r_S}. \end{aligned}$$

Proof. Define $\Omega = \begin{bmatrix} 2cH' & \frac{1}{\beta} I_N \\ \frac{1}{\beta} I_N & \frac{1}{\alpha} I_N \end{bmatrix} \otimes I_n$. It can be known from the condition of Theorem 2 that Ω is positive definite. Select following Lyapunov function

$$V = \frac{1}{2} \bar{e}^T \Omega \bar{e}. \quad (13)$$

Taking the derivative of V gets

$$\begin{aligned}
\dot{V} &= \bar{e}^T \Omega \dot{\bar{e}} \\
&= \bar{e}^T(t) \Omega (M \bar{e}(t) + \bar{F}(x, t)) \\
&= \bar{e}^T(t) \left(\frac{1}{2} \Omega M + \frac{1}{2} M^T \Omega^T \right) \bar{e}(t) + \bar{e}^T(t) \Omega \bar{F}(t) \\
&= \begin{bmatrix} e^{(1)T}(t), & e^{(2)T}(t) \end{bmatrix} \begin{bmatrix} -\frac{\alpha c}{\beta} H' & 0_N \\ 0_N & \frac{1}{\beta} I_N - \frac{\beta c}{\alpha} H' \end{bmatrix} \otimes I_n \begin{bmatrix} e^{(1)}(t) \\ e^{(2)}(t) \end{bmatrix} \\
&\quad - \frac{1}{\beta} e^{(1)T}(t) F(t) - \frac{1}{\alpha} e^{(2)T}(t) F(t) \\
&\leq -\frac{\alpha c}{\beta} \lambda_{\min}(H') e^{(1)T}(t) e^{(1)}(t) + \left(\frac{1}{\beta} I_N - \frac{\beta c}{\alpha} \lambda_{\min}(H') \right) e^{(2)T}(t) e^{(2)}(t) \\
&\quad + \frac{1}{2\beta} e^{(1)T}(t) e^{(1)}(t) + \frac{1}{2\alpha} e^{(2)T}(t) e^{(2)}(t) + \left(\frac{1}{2\beta} + \frac{1}{2\alpha} \right) F(t)^T F(t) \\
&\leq -\frac{\alpha c}{\beta} \lambda_{\min}(H') e^{(1)T}(t) e^{(1)}(t) + \left(\frac{1}{\beta} I_N - \frac{\beta c}{\alpha} \lambda_{\min}(H') \right) e^{(2)T}(t) e^{(2)}(t) \\
&\quad + \frac{1}{2\beta} e^{(1)T}(t) e^{(1)}(t) + \frac{1}{2\alpha} e^{(2)T}(t) e^{(2)}(t) \\
&\quad + \left(\frac{1}{\beta} + \frac{1}{\alpha} \right) \left(\rho_1^2 e^{(1)T}(t) e^{(1)}(t) + \rho_2^2 e^{(2)T}(t) e^{(2)}(t) \right) \\
&= \left(-\frac{\alpha c}{\beta} \lambda_{\min}(H') + \frac{1}{2\beta} + \frac{1}{\beta} \rho_1^2 + \frac{1}{\alpha} \rho_1^2 \right) e^{(1)T}(t) e^{(1)}(t) \\
&\quad + \left(-\frac{\beta c}{\alpha} \lambda_{\min}(H') + \frac{1}{\beta} + \frac{1}{2\alpha} + \frac{1}{\beta} \rho_2^2 + \frac{1}{\alpha} \rho_2^2 \right) e^{(2)T}(t) e^{(2)}(t) \\
&= -\bar{e}^T(t) \begin{bmatrix} m_1 I_N & 0_N \\ 0_N & m_2 I_N \end{bmatrix} \otimes I_n \bar{e}(t) \leq -r V(t),
\end{aligned} \tag{14}$$

where $m_1 = \frac{\alpha c}{\beta} \lambda_{\min}(H') - \frac{1}{2\beta} - \frac{1}{\beta} \rho_1^2 - \frac{1}{\alpha} \rho_1^2$, $m_2 = \frac{\beta c}{\alpha} \lambda_{\min}(H') - \frac{1}{\beta} - \frac{1}{2\alpha} - \frac{1}{\beta} \rho_2^2 - \frac{1}{\alpha} \rho_2^2$, and $r = \frac{2 \min\{m_1, m_2\}}{\lambda_{\max}(\Omega)}$.

When DoS attack occurs, $c = 0$ for $t \in [t_\theta, t_\theta + \tau_\theta)$, then $r = r_D < 0$.

When the DoS attack stops, $c > 0$ for $t \in [t_\theta + \tau_\theta, t_{\theta+1})$, then $r = r_S > 0$.

Therefore,

$$\begin{aligned}
\dot{V} &\leq -r_D V, \text{ when } t \in [t_\theta, t_\theta + \tau_\theta), \\
\dot{V} &\leq -r_S V, \text{ when } t \in [t_\theta + \tau_\theta, t_{\theta+1}).
\end{aligned}$$

Thus,

$$\begin{aligned}
V(t) &\leq e^{-r_S(t-t_\theta-\tau_\theta)} V(t_\theta + \tau_\theta) \\
&\leq e^{-r_D \tau_\theta - r_S(t-t_\theta-\tau_\theta)} V(t_k) \\
&\leq V(t_0) e^{(r_S - r_D)(t-t_0) \frac{1}{T} - r_D \varphi - r_S(t-t_0-\varphi)}.
\end{aligned}$$

According to conditions (a_2) and (b_2)

$$(r_S - r_D) \frac{1}{T} - r_S < 0.$$

In summary, if conditions (a_2) and (b_2) in Theorem 2 are satisfied, the system can achieve group consensus, so Theorem 2 is proved. $\square$

Remark 3. According to Definition 3, we know that the implementation of group consensus has a requirement on the duration of DoS attack. When the controller gain and the attack duration satisfy (a_2) and (b_2) , respectively, group consensus of systems (7) and (8) can be reached with the distributed controller (9).

Remark 4. According to the conditions (a_1) and (a_2) of theorems 1 and 2 respectively, both first-order and second-order MASs subjected to DoS attacks need a certain amount of coupling strength to achieve group consensus. Meanwhile, $c > 0$ also implies that it is crucial for the external controls to achieve group consensus for the MASs subjected to DoS attacks. Base on the conditions (b_1) and (b_2) of theorems 1 and 2 respectively, T should be greater than a certain value when the first-order or second-order MASs subjected to DoS attacks achieve group consensus. In other words, in order to achieve group consensus for the MASs subjected to DoS attacks, the average interval of DoS attacks should be greater than one value. Additionally, the lower bounds of both coupling strength and constant T can be derived from theorems 1 and 2, respectively.

4 Simulation results

In this section, numerical examples are presented to explain the effectiveness of the designed controllers and the obtained theoretical results. Simulations for both first-order MASs and second-order MASs are given.

4.1 Simulations for first-order MASs

A MAS consisting of three leaders and three groups of ten followers are considered, and their dynamics are described by systems (1) and (2). The adjacency matrix of follower agents is as follows:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Based on the adjacency matrix, the topology of MASs, where the same group of followers is represented by the same color, is shown in Fig. 1. Ten follower agents are divided into three subgroups. Especially, agents 1, 2 and 3 are chosen as the followers of the first subgroup, agents 4, 5 and 6 are chosen as the followers of the second subgroup, and agents 7, 8, 9 and 10 are chosen as the followers of the third subgroup.

The nonlinear function is $f(x) = \frac{1}{5}\sqrt{x^2 + 5}$. Obviously, function $f(x)$ is in accord with Assumption 3 with $\rho = 1/5$. In addition, set $B = [1, 0, 0, 0, 1, 0, 0, 0, 0, 2]$, implying that the first follower, fifth follower and tenth follower are pinned by the first leader, second leader and third leader, respectively. Only one agent is pinned in each subgroup.

The initial value of all followers is set in the following vector $x_0 = [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]$, and the initial value of three

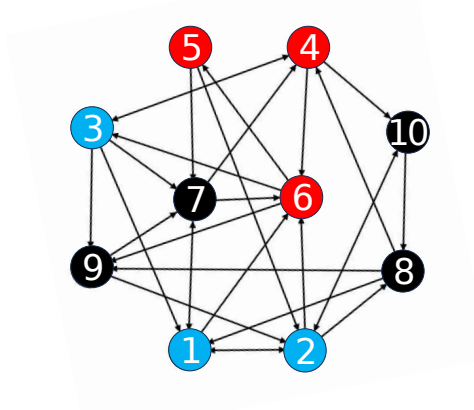


Fig. 1 The topology graph of followers.

leaders x^* is set as $x_0^* = [20, 30, 10]$. By calculating, $\lambda_{\min}(H') = 0.2049$. Set $\alpha = 1$, $c = 3$ and $T = 7$. Therefore, conditions of Theorem 1 are satisfied. The trajectories of position states of both ten followers and three leaders are depicted in Fig. 2, where three thick lines in red, black and blue indicate the trajectories of three leaders. Fig. 2 shows that group consensus is reached. All the followers in each subgroup track the corresponding leader of that subgroup.

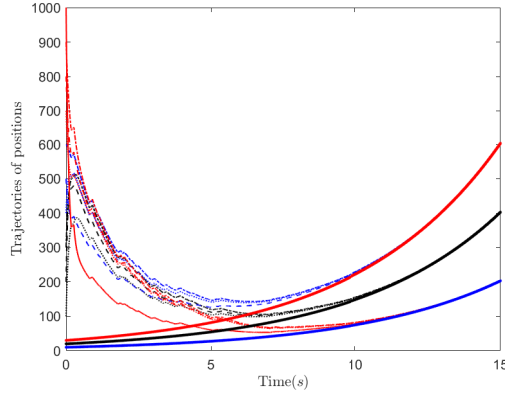


Fig. 2 The evolutions of position states of agents.

In addition, the trajectories of controllers are drawn in Fig. 3, where the MASs are affected by a series of DoS attacks. As can be seen from Fig. 3, when the MASs are affected by the DoS attack, the external control fails and the corresponding external control becomes zero; when a period of DoS attack ends, the communication of the MASs are restored and the external control continues to be effective. Furthermore, the controllers of ten followers gradually tend to 0 as group consensus is asymptotically realized.

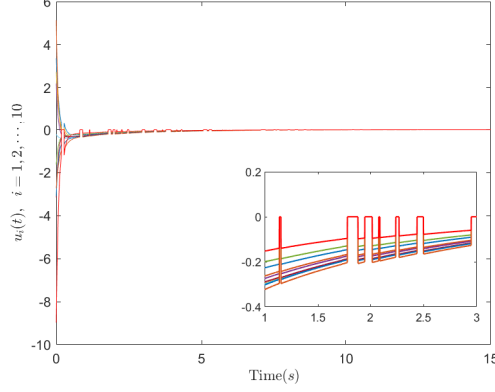


Fig. 3 The evolutions of $u_i(t), i = 1, 2, \dots, 10$.

Remark 5. *Simulation results of first-order MASs show that when three groups of followers of the MASs are controlled via the controller (3), all the follower agents can asymptotically track the corresponding leaders. This also shows that the controller (3) designed in subsection 3.1 is effective for the first-order MASs (1)-(2).*

4.2 Simulations for second-order MASs

In this subsection, a MAS consisting of three leaders and three groups of followers are considered, and their dynamics are described by systems (7) and (8), where $f(x) = \frac{1}{20}\sqrt{x^2 + v^2 + 5}$ with $\rho_1 = 1/20$ and $\rho_2 = 1/20$. The topology of second-order MASs is the same as Fig. 1.

Select $B = [1, 0, 0, 0, 1, 0, 0, 0, 0, 2]$. The initial values of positions and velocities of followers are selected as $x_0 = [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]$, and $v_0 = [50, 100, 150, 200, 250, 300, 350, 400, 450, 500]$, respectively. The initial values of position and velocities of leaders are set as $x_0^* = [20, 40, 10]$, and $v_0^* = [15, 25, 30]$, respectively.

Nodes 1, 2 and 3 are chosen as the first group of followers; Nodes 4, 5 and 6 are chosen as the second group of followers; Nodes 7, 8, 9 and 10 are chosen as the third group of followers. By calculating, $r_S = 0.0029$, $r_D = -0.0327$. Set $\alpha = 1$, $\beta = 1$, $c = 8$ and $T = 15$. Therefore, conditions of Theorem 2 are satisfied. The evolutions of position states and velocity states of all agents are depicted in Fig. 4 and Fig. 5, respectively.

Furthermore, the trajectories of controllers are drawn in Fig. 6. where the MASs are affected by a series of DoS attacks.

Remark 6. *Simulation results of the second-order MASs show that when three groups of followers of the MASs are controlled via the controller (9), all the follower agents can asymptotically track the corresponding leaders. Therefore, the controller (9) is effective for the second-order MASs (7)-(8).*

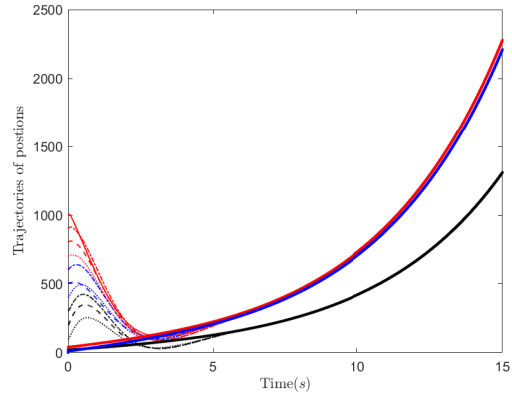


Fig. 4 The evolutions of position states of all agents.

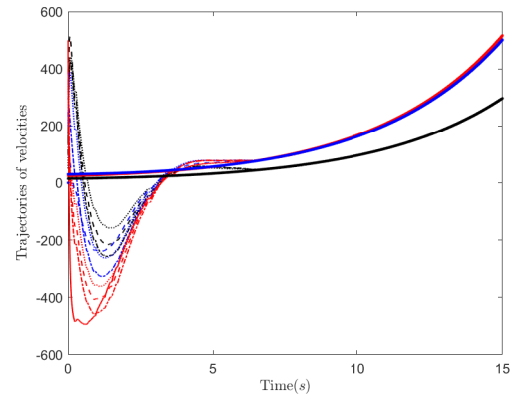


Fig. 5 The evolutions of velocity states of all agents.

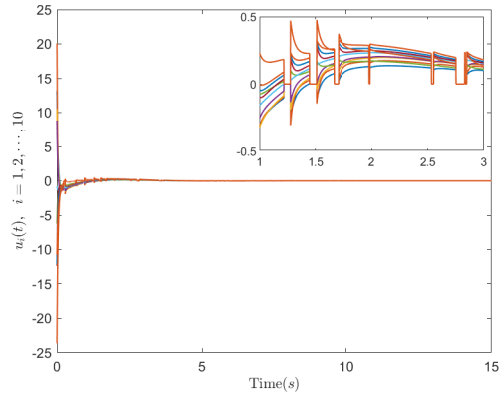


Fig. 6 The evolutions of $u_i(t), i = 1, 2, \dots, 10$.

5 Conclusions

In this article, group consensus of both first-order MASs and second-order MASs under DoS attacks was investigated. According to the method of grouping, the controllers for both first-order nonlinear MASs and second-order nonlinear MASs were designed. By means of the Lyapunov function method, group consensus could be achieved when control gain and durations of DoS attacks satisfied the derived conditions. The designed controller and theoretical results were explained by simulation results. In the future, group consensus of the heterogeneous MASs with switching topology and DoS attacks will be further investigated.

Acknowledgement

This work was supported by the Qing Lan Project of Jiangsu Province of China and the National Natural Science Foundation of China under Grant no. 42375016.

Conflict of interest

The authors declare that they have no conflict of interest.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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